Validation of a finite-element solver for one-dimensional viscous sheet retraction

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August 14, 2026

Abstract

We validate a finite-element solver for the one-dimensional (long-wave) retraction of a viscous liquid sheet against three further implementations of the same model and against the asymptotic results of Ahsan *et al.* [1]. One of the three is independent of this work, a transcription of the code released with that paper; the other two are formulated here, one of them sharing the finite-element framework of the solver under test. Between them they use three different coordinate systems — a Lagrangian mass coordinate, a front-fixed physical grid and a moving mesh — and both finite-element and finite-difference discretisations. Three of them agree to 5.5×10^{-5} in the retraction speed over the whole record; the finite-difference transcription departs by up to 4.5×10^{-3}, but only near the deceleration knee, where its static grid under-resolves the free-edge layer. Measured against the asymptotics, the early-time law and the late-time finite-sheet law are reproduced without adjustable constants, as are the tip-thickness relation and the large-$\mathcal{L}$ crossover time $T_c \to \mathcal{L}$. In the late-time infinite-sheet regime Ahsan *et al.* give the structure $1 - \mathrm{d}X_c/\mathrm{d}T \sim (1+T)^{-2}$ and leave the prefactor undetermined. We measure it. For the canonical rupture state three solvers give $1 - \mathrm{d}X_c/\mathrm{d}T \to \frac{3}{2}(1+T)^{-2}$, stable to 2×10^{-5} under variation of the fit window. The value depends on the initial condition, so 3/2 belongs to that rupture state and is not a universal constant of the model.

1 The model, and where the Ohnesorge number goes

All four solvers integrate one model, which is set out here before them. The physical problem is a planar Newtonian sheet of density ρ, viscosity μ and surface tension γ, initially at rest, of uniform half-thickness h_0 and length l_0, which ruptures at one end at $t = 0$ and retracts. Following the convention of Ahsan *et al.* [1], $h(x,t)$ denotes the *half*-thickness, so the liquid occupies $-h < y < h$; §2.2 maps this onto the full-thickness convention used by the solver under validation.

1.1 The thin-film equations in the viscous limit

For a slender sheet, $|\partial h/\partial x| \ll 1$, the leading-order flow is purely axial and uniform across the thickness, $\mathbf{v} \approx v(x,t)\hat{e}_x$. This is the Trouton approximation [2], and the uniformity is a consequence of *both* surfaces being traction-free; it is what distinguishes a free sheet from a film on a substrate, where no-slip imposes a profile across the layer instead. Appendix A carries out the reduction from the two-dimensional equations. The result is a continuity equation and a

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momentum equation [3, 4],

$$\frac{\partial h}{\partial t} + \frac{\partial}{\partial x}(vh) = 0, \tag{1}$$

$$\underbrace{\rho h \frac{\mathrm{D}v}{\mathrm{D}t}}_{\text{inertial}} = \underbrace{4\mu \frac{\partial}{\partial x}\left(h \frac{\partial v}{\partial x}\right)}_{\text{viscous}} + \underbrace{\gamma h \frac{\partial^3 h}{\partial x^3}}_{\text{capillary}}, \tag{2}$$

with $\mathrm{D}/\mathrm{D}t = \partial_t + v\partial_x$. The factor 4μ is the planar extensional viscosity; the dimensionless Trouton ratio for this geometry is 4.

The retraction is driven at the free edge. A circular tip cannot be imposed there, because $h \propto (x_c - x)^{1/2}$ gives a slope that diverges as $x \to x_c$ and so violates the approximation that produced (1)–(2). Instead Ahsan *et al.* [1] split the domain into a long thin-film region and a small tip region of extent $\mathcal{O}(h)$ in which the full two-dimensional Stokes problem is solved, and match the two [1, 5]. The matching returns an effective boundary condition for the thin-film region,

$$4\mu\, h(x_c, t)\, \frac{\partial v}{\partial x}(x_c, t) = -\gamma, \tag{3}$$

a balance between the viscous force transmitted down the sheet and the capillary force on the tip. Its derivation assumes the tip region is inertialess, which requires $\mathrm{Oh}_p \gg 1$ and, separately, that the elapsed time is not asymptotically large compared with a viscous timescale — a restriction that matters for §8 and is treated in Appendix A of [1].

Two consequences follow at large Ohnesorge number $\mathrm{Oh}_p = \mu/\sqrt{\rho h_0 \gamma}$, which is built on the half-thickness and is the Ohnesorge number of Ahsan *et al.* [1]; §2.2 relates it to the full-thickness Oh used elsewhere here, $\mathrm{Oh}_p = 2\,Oh$. First, (3) fixes the scale of the flow: with h^*, v^* and l^* the characteristic thickness, velocity and axial length, $4\mu h^* v^*/l^* \sim \gamma$. Comparing the two right-hand terms of (2) then gives

$$\frac{\text{capillary}}{\text{viscous}} \sim \left(\frac{h^*}{l^*}\right)^2 \ll 1, \tag{4}$$

so capillary stress is negligible *in the bulk* and survives only through the edge condition (3). Surface tension still drives the whole problem; it simply does so from the tip rather than from the interior.

Second, inertia cannot be discarded, however viscous the liquid. Appendix A.3 shows that setting the left-hand side of (2) to zero and integrating against (3) forces a spatially uniform viscous stress, which is incompatible with a sheet that is initially flat and at rest. Inertia is what keeps the problem well posed, so "large Oh_p" here means *capillary stresses leave the bulk*, not *Stokes flow*.

1.2 The normalisation that removes the Ohnesorge number

Retaining the inertial and viscous terms of (2) and balancing them against (3) yields the visco-inertial length and time [1],

$$l_I = 4\mu\sqrt{\frac{h_0}{\rho\gamma}} = 4h_0\,\mathrm{Oh}_p, \qquad t_I = \frac{4\mu h_0}{\gamma}, \qquad \frac{l_I}{t_I} = u_{TC} = \sqrt{\frac{\gamma}{\rho h_0}}. \tag{5}$$

Note that $l_I \gg h_0$ precisely because $\mathrm{Oh}_p \gg 1$, so the inertial adjustment occurs over distances long compared with the thickness and the thin-film approximation remains self-consistent. Scaling on these,

$$X = \frac{x}{l_I}, \quad T = \frac{t}{t_I}, \quad X_c = \frac{x_c}{l_I}, \quad H = \frac{h}{h_0}, \quad V = \frac{v}{u_{TC}}, \tag{6}$$

and dropping the capillary term by (4), the model becomes

$$\frac{\partial H}{\partial T} + \frac{\partial}{\partial X}(HV) = 0, \qquad H\frac{\mathrm{D}V}{\mathrm{D}T} = \frac{\partial}{\partial X}\left(H\frac{\partial V}{\partial X}\right), \tag{7}$$

with $H(X_c,T)\,\partial_X V(X_c,T) = -1$ and $V(X_c,T) = \mathrm{d}X_c/\mathrm{d}T$ at the free edge, $V = \partial_X H = 0$ at the fixed end, and $H(X,0) = 1$, $V(X,0) = 0$, $X_c(0) = 0$. The domain is $X_c(T) < X < \mathcal{L}$, so this is a free-boundary problem.

No parameter appears in (7). Viscosity has been absorbed into the units, and with it the Ohnesorge number: μ enters both l_I and t_I, and the combination that survives in (7) is dimensionless and equal to one. This is why a single solution serves every viscosity, and it is the reason the curves in this document carry no Oh label.

The Ohnesorge number does not vanish from the problem, however. It reappears in exactly one place — the length of the sheet measured in the new unit,

$$\mathcal{L} = \frac{l_0}{l_I} = \frac{l_0}{4h_0\,\mathrm{Oh}_p}, \tag{8}$$

which enters only through the position of the fixed end. $\mathcal{L}$ is the single dimensionless group of the problem. Because l_0/h_0 and Oh_p are both large and independent, $\mathcal{L}$ may take any value, and the semi-infinite sheet is the limit $\mathcal{L} \to \infty$.

1.3 The exact collapse, and what it is not

The system (7) admits a conserved quantity: $V + H^{-1}\partial_X H$ is material, and the rest state fixes its value at zero, so [1]

$$V = -\frac{1}{H}\frac{\partial H}{\partial X}. \tag{9}$$

Substituting (9) into the continuity equation reduces the coupled pair to a single linear equation for the thickness,

$$\frac{\partial H}{\partial T} = \frac{\partial^2 H}{\partial X^2}, \tag{10}$$

with the edge condition becoming $\mathrm{D}H/\mathrm{D}T(X_c,T) = 1$, hence $H(X_c,T) = 1+T$: the tip thickens linearly in time.

Here, we note that (10) is not a"reduced model". The heat equation is an *exact* consequence of (7), obtained by integrating the conservation law (9); it is not the inertialess limit of anything. Inertia has not been dropped — it has been integrated out. A solver that advances (10) and a solver that advances (7) directly are therefore solving the same physical model by different routes, and agreement between them tests discretisation rather than modelling. That distinction is what makes the comparisons in this document meaningful, and it is recorded in Table 1.

2 Scope and nomenclature

Two solvers are compared throughout.

TC1D-FEM is the solver under validation: a finite-element discretisation of the long-wave equations in a Lagrangian mass cooardinate, with the sheet thickness h on a cell-constant space and the velocity u on continuous piecewise-linear elements, integrated with BDF2. The free edge is a material point of the mesh, so no interface tracking or remeshing is required.

ABDS-CN is an independent implementation: a transcription into Python of the finite-difference solver released with Ahsan *et al.* [1], which advances a Crank–Nicolson discretisation on a static grid with Picard iteration and a tridiagonal solve. Appendix B establishes that the transcription reproduces the original code to the precision at which the original can be read.

Two further solvers appear: TC1D-MASS, introduced in §5 to test one result independently, and TC1D-ALE, which solves the same model as TC1D-FEM but on a moving mesh in the physical coordinate. Table 1 sets out how the four differ.

The asymptotic predictions of Ahsan *et al.* [1] are treated here as hypotheses to be tested, and are plotted as such: they are not used to normalise or calibrate any solver.

	TC1D-FEM	ABDS-CN	TC1D-MASS	TC1D-ALE
Model solved	(7) directly, inertia explicit	(10), inertia integrated out	(10) recast, inertia integrated out	(7) directly, inertia explicit
Equations	$h_T = -h^2 u_m$, $u_T = 4\mu\,\partial_m(h^2 u_m)$	front-fixed heat equation of Appendix D of Ahsan *et al.* [1]	$H_T = H^2 H_{mm}$	Eulerian form on $[x_c(t), \ell_0]$ with ALE correction
Unknown fields	two (h, u)	two (f, u)	one (H)	two (h, u) plus mesh position
Coordinate	Lagrangian mass	physical, algebraic stretching map	Lagrangian mass	physical, moving mesh
Discretisation	finite element, weak form; h cell-constant, u piecewise linear	finite difference, staggered grid	finite difference, nodal	finite element; h, u and mesh piecewise linear
Time integration	BDF2	Crank–Nicolson with Picard iteration	BDF2, Newton	BDF2
Linear algebra	symbolic Jacobian, sparse direct	tridiagonal (Thomas)	tridiagonal (Thomas)	symbolic Jacobian, sparse direct
Edge condition	natural: $h^2 u_m = -1/(2\mu)$ enters as a weak boundary term	$\surd$ tip shape imposed on the last two nodes	essential: $H(0,T) = 1 + T$	natural closure plus kinematic $\dot{x}_c = u(x_c)$ by Lagrange multiplier
Edge speed obtained as	nodal value of a solved field	one-sided gradient at the front	gradient of a Dirichlet solution	nodal value at the moving edge
Far field	$u = 0$, h free	clamped, Neumann thickness	$H = 1$	$u = 0$, h free

Table 1: The four solvers. All four integrate the same physical model, §1; they differ in which equivalent form of it they advance and in the discretisation. TC1D-FEM and TC1D-MASS share the mass coordinate but differ in the equation integrated, in the number of unknowns, and in whether the edge condition is natural or essential; ABDS-CN works in a physical coordinate and so covers the case those two do not. TC1D-ALE uses the same model and framework as TC1D-FEM but places the sheet in the physical coordinate on a moving mesh, which is the discretisation the two-dimensional solver uses; it therefore tests that machinery on a problem with a known answer.

2.1 How closely the four agree

Because all four integrate the same model, disagreement between them is numerical error and nothing else. Taking TC1D-FEM as the reference and comparing by spline, so that the comparison itself contributes nothing, the worst relative difference in $\mathrm{d}X_c/\mathrm{d}T$ over the whole record is

$\mathcal{L}$	TC1D-MASS	TC1D-ALE	ALE vs MASS	ABDS-CN
1	7.7×10^{-7}	4.4×10^{-6}	4.5×10^{-6}	3.9×10^{-3}
5	4.7×10^{-6}	1.0×10^{-5}	5.7×10^{-6}	3.0×10^{-4}
20	2.6×10^{-5}	3.5×10^{-5}	8.8×10^{-6}	2.5×10^{-4}
40	3.7×10^{-5}	5.5×10^{-5}	1.9×10^{-5}	4.5×10^{-3}
200	3.4×10^{-6}	3.3×10^{-6}	7.9×10^{-7}	4.6×10^{-4}

Three of the four agree to 5.5×10^{-5} everywhere. ABDS-CN is the outlier, and only near the deceleration knee at $T \approx \mathcal{L}$, where its largest departures occur. The mechanism is its static grid: the physical spacing is $\mathcal{L}/N$, so the $\mathcal{O}(\sqrt{T})$ edge layer is progressively under-resolved as $\mathcal{L}$ grows at fixed N. The three solvers that track the edge, two on a mass coordinate and one on a moving mesh, bracket each other to 5.5×10^{-5} through the same knee. This does not change the conclusion of §5; it sharpens it, because the coefficient measured there now rests on three codes that are mutually converged rather than on one code and a reference.

One practical note for anyone reproducing these numbers. The solvers write on different geometric output schedules, so comparing them requires interpolation. Linear interpolation across the knee manufactures up to 7×10^{-3} of apparent disagreement — two orders of magnitude larger than the real differences tabulated above — between curves that in fact agree to 5×10^{-5}. The figures therefore plot each solver on its own abscissa, and the table above is computed by spline.

2.2 Units

Ahsan *et al.* [1] scale on the half-thickness $h_0 = H/2$, with $u_{TC} = \sqrt{\gamma/\rho h_0}$, Ohnesorge number $Oh_p = \mu/\sqrt{\rho h_0 \gamma}$ and visco-inertial time $t_I = 4\mu h_0/\gamma$. The present formulation scales on the full thickness H, with the Savva and Bush [6] Ohnesorge number Oh and viscous time $\tau_{\text{vis}} = \mu H/2\gamma$. The maps are

$$Oh_p = 2\,Oh, \qquad T = t^*/4, \qquad \mathcal{L} = \frac{\ell_0}{4h_0 Oh_p} = \frac{M}{4\,Oh}, \tag{11}$$

where $t^* = t/\tau_{\text{vis}}$ and M is the total sheet mass per unit width, which is the extent of the computational domain in the mass coordinate. The retraction speed of Ahsan *et al.* [1], $\mathrm{d}X_c/\mathrm{d}T$, equals $u^* = U/V_{TC}$ with no intervening factor. These relations are confirmed by two physical identities: the tip condition $HV_X = -1$ is the natural boundary condition of the present formulation, and $H(X_c, T) = 1 + T$ is the tip-thickness relation measured here as $h_{\text{tip}} = 1 + t^*/4$.

Equation (11) also shows that the present formulation represents a sheet of finite length: $\mathcal{L}$ is set by the mass domain M, and the far boundary condition (vanishing velocity, free thickness) is the condition imposed by Ahsan *et al.* [1]. The mapping was checked before use: at $M = 20$, $Oh = 1$, the deceleration is observed at $T_c = 4.97$ against $\mathcal{L} = 5$.

3 Early time: $T \ll 1$

At early times Ahsan *et al.* [1] predict a diffusive growth of the retraction speed,

$$\frac{\mathrm{d}X_c}{\mathrm{d}T} \sim 2\sqrt{T/\pi}, \tag{12}$$

Figure 1 compares both solvers and (12) at $\mathcal{L} = 0.1$, 1 and 20.

Equation (12) is reproduced to better than 0.06% for $T \leq 10^{-3}$ at every $\mathcal{L}$. Its range of validity depends strongly on $\mathcal{L}$: taking the time at which the relative deviation first exceeds 1%, 5% and 10%,

	1%	5%	10%
$\mathcal{L} = 0.1$	$T = 0.0031$	0.0053	0.0075
$\mathcal{L} = 1$	$T = 0.019$	0.102	0.213
$\mathcal{L} = 20$	$T = 0.019$	0.102	0.221

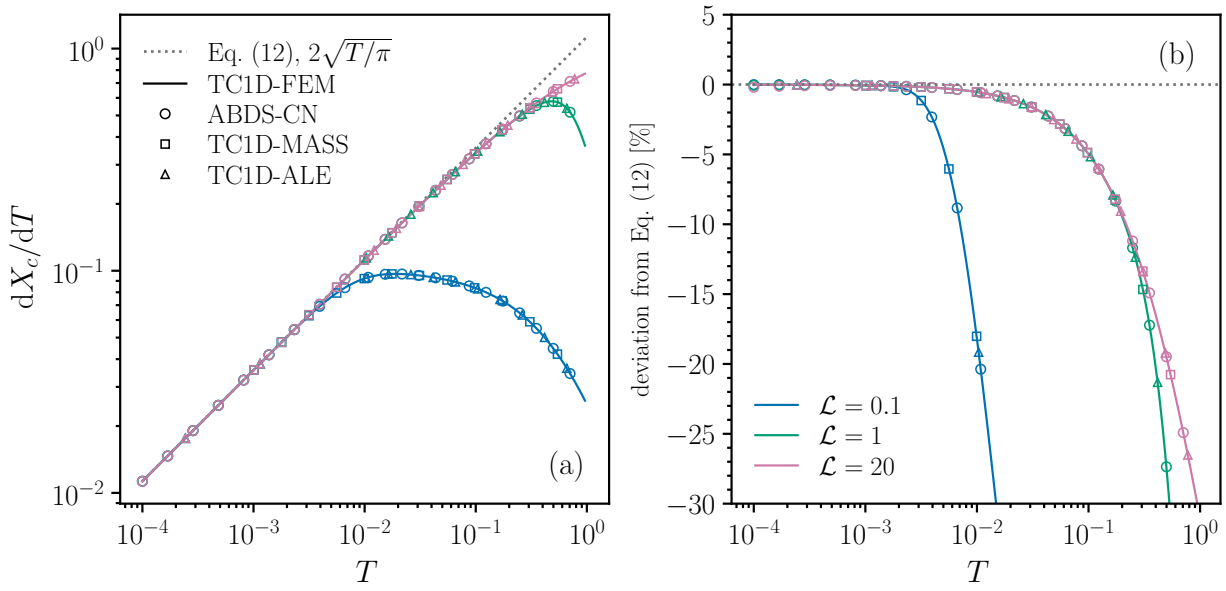


Figure 1: Retraction speed at early time, all four solvers. Colour encodes $\mathcal{L}$ and is keyed in (b); marker encodes the solver and is keyed in (a). Solid: TC1D-FEM. Circles: ABDS-CN. Squares: TC1D-MASS. Triangles: TC1D-ALE. Dotted: (12). The four agree to 3×10^{-4} over the whole window. TC1D-MASS markers begin at $T = 10^{-3}$, the first time on its geometric output schedule.

The two solvers agree over this window to a maximum absolute difference of 2.4×10^{-5} at $\mathcal{L} = 0.1$, 1.3×10^{-4} at $\mathcal{L} = 1$ and 2.1×10^{-5} at $\mathcal{L} = 20$; the largest of these, 1.3×10^{-4}, is the figure quoted in the abstract.

4 Late time at finite length: the deceleration

When the free edge approaches the far end of the sheet, $\mathcal{L} - X_c \ll 1$, Ahsan *et al.* [1] predict $\mathrm{d}X_c/\mathrm{d}T \sim \mathcal{L}(1+T)^{-2}$ (eq. 6.10). This regime is the principal focus of that paper.

The prediction is reproduced without any adjustable constant. At the end of each run the ratio of the computed speed to eq. (6.10) of Ahsan *et al.* [1] lies between 1.00002 and 1.0010, and the remaining sheet length $\mathcal{L} - X_c$ agrees with $\mathcal{L}/(1+T)$ to 1.3×10^{-4}. Continuing the $\mathcal{L} = 5$ case to $T = 400$ shows the relative departure decaying as $O(T^{-2})$, which is the order of the neglected terms.

5 Late time at large length

The regime of principal interest here is $\mathcal{L} \gg T \gg 1$, in which the edge has reached the Taylor–Culick speed but has not yet sensed the far end. Ahsan *et al.* [1] give, for an infinite sheet,

$$\frac{\mathrm{d}X_c}{\mathrm{d}T} \sim 1 - \frac{1}{(1+T)^2}. \tag{13}$$

Two properties of eq. (13) govern how it can be tested. The deficit is $(1+T)^{-2}$, not a pure power T^{-2}. Fitting a power law in T therefore tests only the leading late-time exponent, and not the unit shift or the prefactor, which is where the content of eq. (13) lies. The coefficient is described by Ahsan *et al.* [1] as a numerical fit, with the corresponding expansion coefficient stated to remain undetermined by the asymptotic analysis and to depend on the history of the retraction.

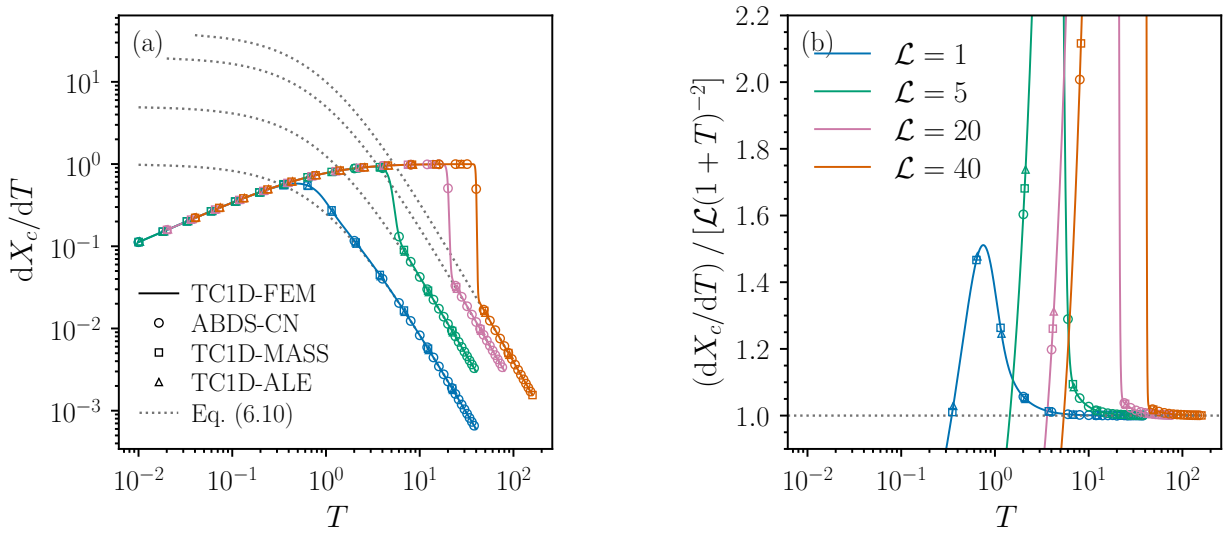


Figure 2: Deceleration at finite $\mathcal{L}$, all four solvers. Colour encodes $\mathcal{L}$ and is keyed in (b); line style encodes the solver and is keyed in (a). Solid: TC1D-FEM. Circles: ABDS-CN. Squares: TC1D-MASS with the fixed-end condition of §A.3. Triangles: TC1D-ALE. Dotted: $\mathcal{L}(1+T)^{-2}$. Markers are drawn sparsely and at staggered phase because the four curves are indistinguishable at this scale; the differences are quantified in §2.1.

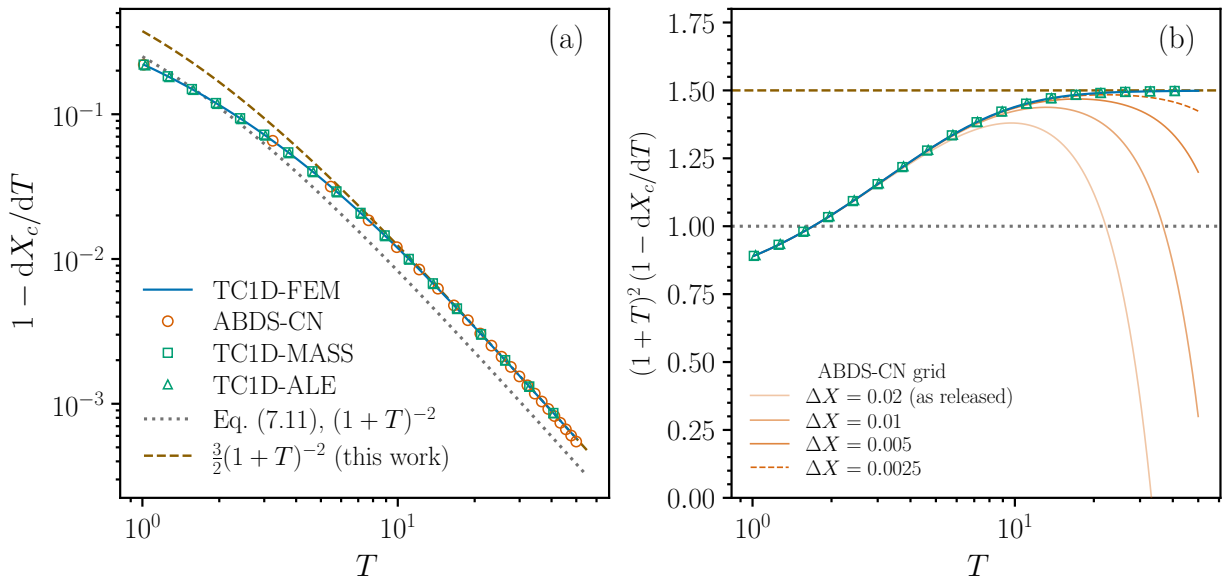


Figure 3: Late-time speed and deficit at large $\mathcal{L}$. Solid: TC1D-FEM. Circles: ABDS-CN at its resolved grid. Squares: TC1D-MASS. Triangles: TC1D-ALE at $\mathcal{L} = 200$. Dotted: eq. (13). In (b) the three solvers that resolve the edge layer approach 3/2 together, while the ABDS-CN curves fall away in order of grid spacing; the coefficient of unity is what that spacing produces, not what the model does.

5.1 The measured deficit

Figure 3 shows the speed and the deficit $1 - \mathrm{d}X_c/\mathrm{d}T$. The ratio

$$R(T) = (1+T)^2\Big(1 - \frac{\mathrm{d}X_c}{\mathrm{d}T}\Big), \tag{14}$$

has an intercept that the $\sim$ in (13) does not fix. The three solvers give

solver	model solved	$R(50)$
TC1D-FEM (finite element, mass coordinate)	momentum form	1.4986
ABDS-CN (finite difference, Richardson $\Delta X \to 0$)	reduced form	1.4985
TC1D-MASS (nonlinear diffusion, fixed domain)	$H_T = H^2 H_{mm}$	1.4986

In the mass coordinate the exact invariant reduces the problem to a nonlinear diffusion equation on a fixed domain, in which the free boundary disappears and the tip stress condition becomes an ordinary Dirichlet condition $H(0,T) = 1+T$. It agrees with the other two to six digits.

Continued to $T = 500$ at four timestep levels, with an observed convergence order of 1.993–1.998 throughout, the Richardson-extrapolated ratio is

T	50	100	200	300	400	500
R	1.498623	1.499660	1.499919	1.499993	1.500132	1.500319

R rises monotonically towards 3/2 and does not turn down. Up to $T \approx 300$ the residual $3/2 - R$ is positive and decays as $(1+T)^{-2}$; beyond that the tabulated values overshoot by a few parts in 10^4, which is the size of the timestep extrapolation there rather than a feature of the limit (the finest computed value at $T = 500$ reads 1.624 before extrapolation). We therefore quote the limit from the fit below rather than from the largest T reached. Writing $s = (1+T)^{-1}$ and $R = a_2 + a_3 s + a_4 s^2 + \ldots$, so that a_2 is the $s \to 0$ intercept, polynomial fits of degree one to three with the window start slid outward give an intercept stable at 1.50000, varying by less than 2×10^{-5} across window starts $T_{\min} = 50$ to 125. We take

$$\lim_{T\to\infty} (1+T)^2\Big(1 - \frac{\mathrm{d}X_c}{\mathrm{d}T}\Big) = 1.5000 \pm 0.0005, \tag{15}$$

i.e. 3/2 to within the quoted uncertainty. We report this as a measurement for the canonical rupture state rather than as an exact or universal value; §5.4 shows it is a property of the initial condition. The same fits give $a_3 = 0.000 \pm 0.001$: there is no $(1+T)^{-3}$ term, so $(1+T)$ is the exact grouping, $U_{-3} = -2U_{-2} = 3$, and the next correction enters at $O\big((1+T)^{-4}\big)$.

The result is insensitive to every numerical parameter varied independently: mesh spacing (a factor of eight, changing $R(50)$ by 10^{-6}), domain truncation ($M = 70$, 90, 150, identical to every printed digit), timestep (two different step-control strategies, extrapolated limits differing by 3×10^{-5}), and, for ABDS-CN, the truncation length and ΔX.

Two further measurements of the same constant avoid the tip gradient entirely. From the displacement, $(X_c - T + 1)(1+T)$ measures 1.4995 at $T = 50$; from the shape of the first correction to the thickness profile, away from the tip, 1.51 at $T = 50$.

5.2 The recommended form, and one that should be avoided

We therefore report

$$\frac{\mathrm{d}X_c}{\mathrm{d}T} = 1 - \frac{3}{2}\frac{1}{(1+T)^2} + O\big((1+T)^{-4}\big), \qquad X_c = T - 1 + \frac{3}{2}\frac{1}{1+T} + O(T^{-2}). \tag{16}$$

The single term is accurate to 0.09% over $50 \le T \le 200$ and 0.02% over $100 \le T \le 200$. Adding a $(1+T)^{-3}$ term to enforce the integral constraint makes the fit *worse* by more than an order of magnitude at large T, because it inserts a correction at an order where the data gives none; we recommend against it.

The displacement relation in eq. (16) deserves separate emphasis, because its constant is derivable and independent of the deficit coefficient. Expanding in the exact edge thickness $\theta = 1+T$ gives $X_c = \theta - 2 + a_2/\theta$, whose $O(1)$ term is -1 for *any* a_2. Measurements give $T - X_c = 0.99255$ at $T = 200$ against a predicted 0.99254.

5.3 The integral constraint does not fix the coefficient

The integral constraint $\int_0^\infty (1 - \mathrm{d}X_c/\mathrm{d}T)\,\mathrm{d}T = 1$ is sometimes invoked to fix the coefficient. It cannot: the $O(1)$ term of X_c is -1 regardless of a_2, so the constraint is satisfied identically and carries no information about it. We verify this directly, recovering 1.00002 when the constraint is evaluated with the measured $a_2 = 3/2$.

Equation (13) is written with $\sim$, and the corresponding expansion coefficient is left undetermined. Substituting the uniform-profile approximation (7.13) of Ahsan *et al.* [1] into the integral relation (7.16) does return a prefactor of unity, but that profile reproduces only the leading two terms of the thickness expansion; the same profile in the alternative exact relation (3.11) returns a vanishing deficit. The two routes disagree at the order that would fix a_2, so that order is beyond the approximation. The $\sim$ is the right statement of what the analysis gives.

5.4 Dependence on the initial condition

The coefficient is a property of the initial condition and not a universal constant of the model. Replacing the canonical rupture condition $H \equiv 1$, $V \equiv 0$ with $H(m,0) = 1 + \delta e^{-m}$, which is admissible because the edge condition holds for any initial state, gives

δ	0 (canonical)	0.5	2
a_2	1.4997	1.8229	0.9366

The value 3/2 is therefore the canonical-rupture value. This supports the statement of Ahsan *et al.* [1] that the coefficient depends on the history of the retraction. The grouping $\theta = 1 + T$ and the result $a_3 = 0$ are unchanged across all three initial conditions.

5.5 Profiles and the tip thickness

Figure 4 compares thickness and velocity profiles against eqs. (7.13) and (7.14) of Ahsan *et al.* [1]. The tip-thickness relation $H(X_c, T) = 1 + T$ is reproduced to a maximum relative deviation of 3.6×10^{-5} over $1 \le T \le 50$. The profile predictions agree to 0.043% and 1.55% respectively at $T = 50$, improving monotonically with T.

6 A moving-mesh solver

TC1D-ALE solves the same model as TC1D-FEM, with inertia retained, but in the physical coordinate on a mesh that moves. It exists to test the arbitrary Lagrangian–Eulerian machinery on a problem whose answer is known, because that machinery is what the two-dimensional solver relies on.

The mesh velocity w is neither the material velocity nor zero, so every time derivative taken at a fixed mesh point carries the convective correction $\mathrm{D}/\mathrm{D}t = \partial_t|_{\mathrm{mesh}} + (u - w)\partial_x$. At the free edge $w = u$, so the mesh tracks the interface; in the interior w is the affine stretch $w = u(x_c)(\ell_0 - x)/(\ell_0 - x_c)$, obtained as the solution of a mesh-motion equation rather than by

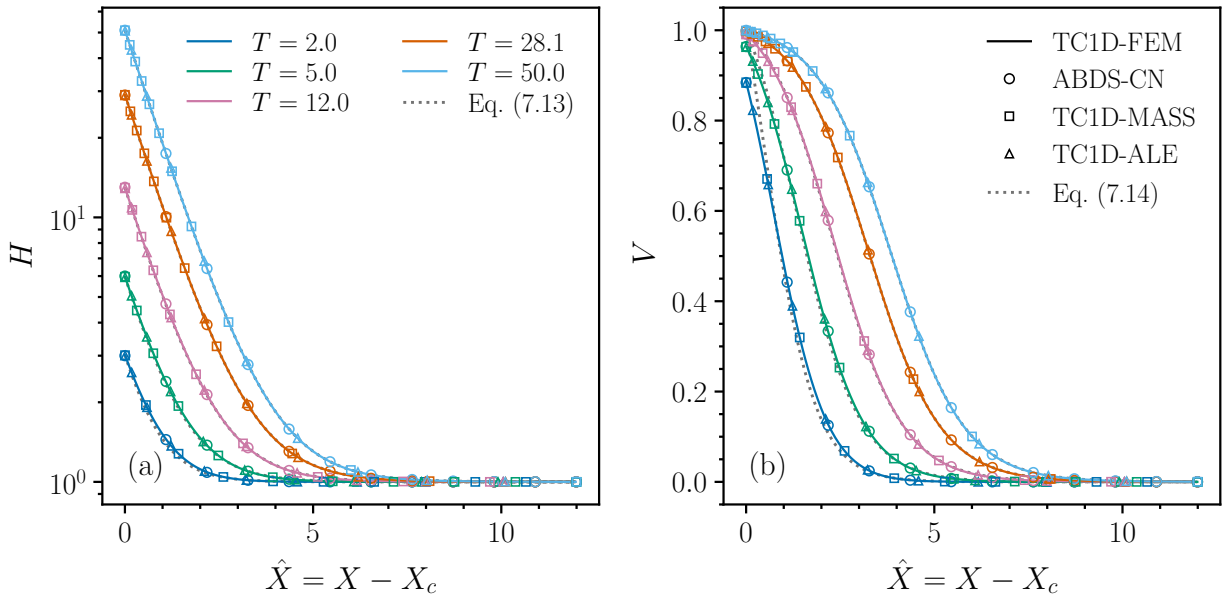


Figure 4: Thickness and velocity profiles at large $\mathcal{L}$, all four solvers. Colour encodes T and is keyed in (a); marker encodes the solver and is keyed in (b). Solid: TC1D-FEM. Circles: ABDS-CN. Squares: TC1D-MASS, mapped out of the mass coordinate by $\hat{X} = \int_0^m \mathrm{d}m'/H$ and $V = -\partial_m H$, both exact relations of the model. Triangles: TC1D-ALE. Dotted: eqs. (7.13) and (7.14) of Ahsan *et al.* [1]. The added solvers were run to land on the exact output times of the reference profiles and agree with them to 2×10^{-4} in both H and V.

renoding between steps. The edge carries the same natural closure as TC1D-FEM together with the kinematic condition $\dot{x}_c = u(x_c)$, imposed through a Lagrange multiplier because the boundary position depends on an unknown.

It reproduces all three comparison quantities: $u^*(30)$ extrapolates to 0.9807279 in time and 0.9807278 in space, agreeing with each other to 1.5×10^{-7} and with TC1D-FEM's limit to 2.9×10^{-7}; the late-time ratio extrapolates to 1.49860, 1.49964 and 1.50000 at $T = 50$, 100 and 200; and the displacement relation holds with a residual of 10^{-7} relative. Observed convergence is second order in both time ($1.92 \to 1.99$) and space (2.000 at every refinement).

Two checks are worth recording. The mesh sits on the affine stretch to 6×10^{-15} of the domain length, so the mesh-motion solution is verified rather than assumed; and a control run with the same moving mesh but the ALE correction omitted drifts the total mass by a factor of sixteen. The correction is not a refinement but the difference between this result and none.

The physical coordinate loses resolution where the mass coordinate gains it: the rim occupies a fixed mass but a physical extent narrowing as $(1 + T)^{-1}$, while the affine stretch compresses the mesh only in proportion to the remaining sheet length. Late-time accuracy is therefore more expensive here than in TC1D-FEM. This is a property of the coordinate rather than of the implementation, and it is the reason Ahsan *et al.* [1] adopt an algebraic stretching map.

7 The crossover time

For large $\mathcal{L}$ the edge decelerates from the Taylor–Culick speed onto the finite-sheet law of §4. Ahsan *et al.* [1] report the crossover time T_c and its duration ΔT_c as functions of $\mathcal{L}$, with $T_c = \mathcal{L}$ for large $\mathcal{L}$.

Neither quantity is given in closed form in that paper, but both are defined unambiguously by the released code, and it is that definition which is used here and applied identically to every

solver. Write the deceleration as the non-negative part of the edge acceleration,

$$A(T) = \max\left(-\frac{\mathrm{d}^2 X_c}{\mathrm{d}T^2},\ 0\right) = \max\left(-\frac{\mathrm{d}V}{\mathrm{d}T},\ 0\right), \tag{17}$$

evaluated with the same centred difference the released code uses. Then

$$T_c = \arg\max_T A(T), \qquad \Delta T_c = 2\,(T_+ - T_-)\,, \qquad T_\pm : A(T_\pm) = \tfrac{1}{2}\max_T A, \tag{18}$$

so T_c is the instant of strongest deceleration and ΔT_c is twice the full width of that peak at half its height. Two details of (18) are inherited deliberately rather than improved upon, because the object is to reproduce the published figure. The first is the factor of two, discussed below. The second is that T_- and T_+ are taken as the first and last *sampled* times at which $A \geq \frac{1}{2}\max A$, found by a scan over the whole record rather than over the interval containing the peak, so any second deceleration episode elsewhere in the record would widen ΔT_c; the contiguous-interval variant is computed alongside and is identical here, confirming the peak is single.

Equation (18) reads T_c and $T_\pm$ off output grid points with no sub-grid interpolation, so ΔT_c is quantised in steps of twice the output spacing. The runs use a fixed number of uniform samples over a window that grows with $\mathcal{L}$, so that quantum grows too — from 0.013 at $\mathcal{L} = 3$ to 0.050 at $\mathcal{L} = 50$, or about 1.7% of ΔT_c. Every value returned by (18) is an exact multiple of it. This is the origin of the small sawtooth in figure 5(b) at large $\mathcal{L}$, including the single backward step near $\mathcal{L} = 47$; it is a property of the definition and the sampling, not of the solvers, and it affects both codes identically because both are read on the same grid. The smooth curve in figure 5(b) locates $T_\pm$ by linear interpolation of A between the bracketing samples and the peak by a parabola through its three highest, which removes the quantisation; the values quoted below are the unmodified (18) ones, so that the comparison with the digitised data is like for like.

Figure 5 compares the solvers against values digitised from Ahsan *et al.* [1].

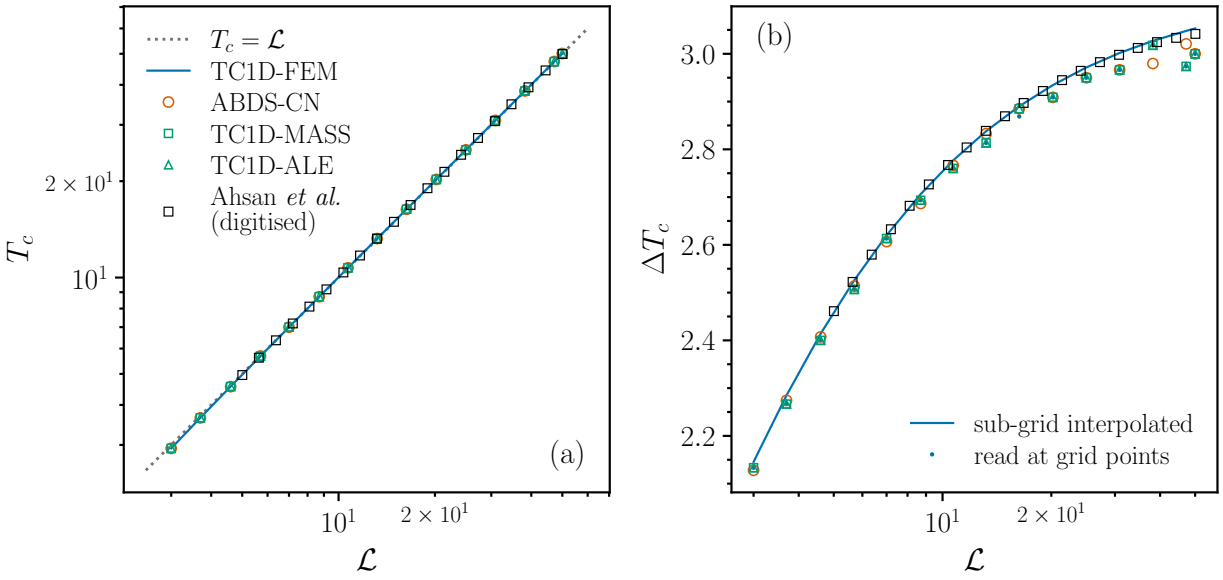


Figure 5: Crossover time T_c and duration ΔT_c against $\mathcal{L}$, by the definition (18). Solid: TC1D-FEM. Circles: ABDS-CN. Green squares: TC1D-MASS. Triangles: TC1D-ALE. Black squares: values digitised from Ahsan *et al.* [1]. The three solvers that track the free edge return the same T_c at every $\mathcal{L}$. In (b) the line is ΔT_c with the half-maximum crossings located by sub-grid interpolation and the dots are the same quantity read at grid points, as (18) does; the dots are quantised in steps of twice the output spacing, and it is that quantisation, not the solvers, which produces the scatter at large $\mathcal{L}$.

We find $T_c/\mathcal{L} = 1$ to within 0.25% for $\mathcal{L} \geq 7$. Against the digitised values, T_c agrees to 0.142% RMS, within the estimated digitisation uncertainty of 0.2%, and ΔT_c to 0.024 absolute (1.5%).

One point of definition should be recorded. The text of Ahsan *et al.* [1] describes ΔT_c as a full width at half maximum, while the released code computes twice that width. The values in the paper are consistent with the code. Both conventions are reported here.

8 The range of validity assumed throughout

The four regimes examined above — the early response of §3, the terminal approach $\mathcal{L} - X_c \ll 1$ at arbitrary $\mathcal{L}$ of §4, the semi-infinite limit $\mathcal{L} \gg T \gg 1$ of §5, and the crossover at finite large $\mathcal{L}$ with $1 < T < \mathcal{L}$ of §7 — share one assumption that is absent from the equations actually solved.

Section 1.2 showed that the Ohnesorge number is absorbed into the units (5), which is what makes the curves of the preceding sections universal and lets a single solution serve every viscosity. It does not follow that the results hold at every viscosity. The reduction remains faithful to the underlying flow only while the thin-film description does, and Ahsan *et al.* [1] bound that by $T = O(Oh_p)$, beyond which they identify a two-dimensional inertial regime governed by the full Navier–Stokes equations. The edge condition (3) carries a restriction of the same kind, since its derivation requires the elapsed time not to be asymptotically large relative to a viscous timescale [1]. Since the deceleration of a sheet of length $\mathcal{L}$ occurs at $T_c \to \mathcal{L}$, placing that crossover inside the thin-film horizon requires

$$\mathcal{L} \ll Oh_p, \tag{19}$$

which in the units of (11) reads $M \ll 8\,Oh^2$. Condition (19) does not make the description valid over the sheet's whole history: the terminal law of §4 is a $T \to \infty$ limit of the reduced model and lies outside any finite horizon, so it is a statement about that model rather than about a real sheet at late time. What (19) secures is the part of the history in which the crossover happens. For a semi-infinite sheet $\mathcal{L} \to \infty$ and no finite viscosity satisfies it at all; what survives is the horizon $T \ll Oh_p$.

This bounds what the present validation can claim, and the boundary is worth being precise about. The agreement documented in §§3–7 is agreement between two discretisations of the same reduced model, and it is established to the precision reported there. It is not evidence that the reduced model describes a sheet at any particular Ohnesorge number, and no such claim is made. Establishing that requires a solver which never performs the reduction.

The question is therefore well posed: for a semi-infinite sheet, how large must Oh_p be before a genuine two-dimensional flow spends a resolvable part of its history in the regime these sections describe? It cannot be answered from within the reduced model, which has already discarded the parameter the answer depends on.

Direct simulation of the two-dimensional problem at $0.77 \le Oh_p \le 10$ indicates that the reduced description is accurate at early times — the local exponent of the deficit agrees with the present solvers to better than 0.01 out to $T \approx 0.5$ at $Oh_p = 10$ — but that the time at which it departs grows only as $Oh_p^{1/2}$, rather than linearly as (19) would suggest. On that scaling a decade of long-wave behaviour would require Oh_p of order 10^2, well above the range in which the long-wave model is usually invoked. Those measurements are reported separately; they are noted here only to mark the boundary of what the present validation establishes.

A From the two-dimensional equations to the thin-film model

This appendix records the reduction used in §1, first retaining inertia and then discarding it. The second case is included not because it is used, but because discarding inertia is the natural thing to try at large Ohnesorge number, and seeing precisely how it fails is what justifies the visco-inertial scaling (5). The systematic versions of the first derivation are due to Erneux and Davis [3] and Howell [4], following Trouton [2]; the treatment here is condensed to the terms that matter for the retraction problem. The matched-asymptotic treatment of the tip region is that of Ahsan *et al.* [1].

A.1 The two-dimensional problem

Take a Newtonian liquid of constant ρ and μ, occupying $-h(x,t) < y < h(x,t)$, with velocity (u, w) and pressure p, surrounded by a passive ambient at zero stress. The sheet is symmetric about $y = 0$, so u is even and w odd in y. The equations are

$$\frac{\partial u}{\partial x} + \frac{\partial w}{\partial y} = 0, \qquad \rho \frac{\mathrm{D}u_i}{\mathrm{D}t} = \frac{\partial \sigma_{ij}}{\partial x_j}, \qquad \sigma_{ij} = -p\,\delta_{ij} + \mu\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right), \tag{20}$$

subject on $y = h(x,t)$ to the kinematic condition and the stress condition

$$\frac{\partial h}{\partial t} + u\frac{\partial h}{\partial x} = w, \qquad \sigma_{ij} n_j = -\gamma\kappa\, n_i, \tag{21}$$

with $\mathbf{n}$ the outward unit normal and $\kappa = \nabla\cdot\mathbf{n}$ the curvature, positive on a bulge. The sign is fixed by the static check: for a quiescent drop $\sigma_{ij}n_j = -p\,n_i$, so (21) gives $p = \gamma\kappa > 0$ inside, as it must be.

A.2 Reduction with arbitrary inertia

Let $\varepsilon = h_0/l_0 \ll 1$ and scale $x \sim l_0$, $y \sim h_0$, $u \sim U$, $w \sim \varepsilon U$, so that continuity balances. Expanding in ε^2,

$$u = u_0(x,t) + \varepsilon^2 u_2(x,y,t) + \ldots, \qquad p = p_0(x,t) + \ldots \tag{22}$$

The transverse momentum equation gives $\partial p/\partial y = \mathcal{O}(\varepsilon^2)$, so the pressure is uniform across the thickness at leading order. The axial equation at leading order reduces to $\mu\,\partial^2 u_0/\partial y^2 = 0$, and the tangential stress condition on *both* free surfaces forces the constant of integration to vanish, leaving u_0 independent of y. This is the step that distinguishes a free sheet from a film on a substrate: with two traction-free surfaces the leading-order profile is plug flow, whereas no-slip on a wall would give a parabolic profile and a lubrication model instead. Continuity then fixes the transverse velocity, $w = -y\,\partial_x u_0$.

The normal stress condition on $y = h$ balances the transverse deviatoric stress and the pressure against the Laplace pressure. For a weakly curved surface the curvature is $\kappa = -\partial^2 h/\partial x^2$, positive on a bulge, so with a passive ambient at zero pressure

$$p_0 = -2\mu\frac{\partial u_0}{\partial x} - \gamma\frac{\partial^2 h}{\partial x^2} + \mathcal{O}(\varepsilon^2), \tag{23}$$

where the first term follows from $\partial w/\partial y = -\partial u_0/\partial x$. The axial stress carried by the sheet is then

$$\sigma_{xx} = -p_0 + 2\mu\frac{\partial u_0}{\partial x} = 4\mu\frac{\partial u_0}{\partial x} + \gamma\frac{\partial^2 h}{\partial x^2}. \tag{24}$$

The factor 4μ is the planar extensional viscosity of Trouton [2]: a factor 2 from the deviatoric stress and a further factor 2 because the transverse contraction $\partial_y w = -\partial_x u_0$ feeds back through the pressure.

Integrating continuity across the thickness and applying the kinematic condition at $y = \pm h$ gives (1) exactly. For momentum, balance the forces on a slice. The mass per unit length is $2\rho h$, and the total axial force transmitted across a section is the depth integral of (24) plus the axial projection of the tension carried by the two surfaces,

$$N = \int_{-h}^{h} \sigma_{xx}\,\mathrm{d}y + \frac{2\gamma}{\sqrt{1+h_x^2}} = 8\mu\,h\frac{\partial u_0}{\partial x} + 2\gamma h\frac{\partial^2 h}{\partial x^2} + 2\gamma - \gamma h_x^2 + \ldots, \tag{25}$$

the projection factor being the only place the surface slope enters at this order. The two capillary contributions in (25) do not act separately: differentiating,

$$\frac{\partial}{\partial x}\left(2\gamma h\,h_{xx} - \gamma h_x^2\right) = 2\gamma h_x h_{xx} + 2\gamma h\,h_{xxx} - 2\gamma h_x h_{xx} = 2\gamma h\,h_{xxx}, \tag{26}$$

so the $h_x h_{xx}$ terms cancel exactly and only the third derivative survives. Keeping the depth integral of σ_{xx} while omitting the inclination of the surface along which the tension acts leaves a spurious $\gamma h_x h_{xx}$ and does not reproduce (2). With $2\rho h\,\mathrm{D}u_0/\mathrm{D}t = \partial_x N$ and division by two, the result is (2) [3, 4].

The constant 2γ has zero gradient and so exerts no bulk force; it acts only where the sheet terminates, which is why surface tension enters this problem through a boundary condition. And $\gamma h\,\partial^3 h/\partial x^3$ is the only place bulk capillarity survives, which by (4) is negligible once $\mathrm{Oh}_p \gg 1$.

Equations (1) and (2) are what TC1D-FEM and TC1D-ALE integrate. In the Lagrangian mass coordinate m with $\mathrm{d}m = h\,\mathrm{d}x$, so that $\partial_x = h\,\partial_m$ and the free edge is a material point, they take the form quoted in Table 1,

$$\left.\frac{\partial h}{\partial t}\right|_m = -h^2\frac{\partial u}{\partial m}, \qquad \rho\left.\frac{\partial u}{\partial t}\right|_m = 4\mu\frac{\partial}{\partial m}\left(h^2\frac{\partial u}{\partial m}\right), \tag{27}$$

in dimensional time, with (3) entering as a natural boundary term. (In the units of (6) the coefficients are unity; the solvers work in code units with $\rho = \gamma = 1$.)

The partial differential equations in (27) are invariant under the swap between half- and full-thickness conventions; the *edge constant* is not. In the half-thickness variables of this appendix (3) reads $h^2\partial_m u = -\gamma/4\mu$, whereas Table 1, which follows the solver's full-thickness convention $h_f = 2h$ and hence $m_f = 2m$, records $h_f^2\partial_{m_f} u = -\gamma/2\mu$. Both are the same physical condition. Only one of them is right for a given code.

A.3 Reduction with vanishing inertia, and why it fails here

Now suppose inertia is negligible, so that the two-dimensional problem is unsteady Stokes flow: $\partial\sigma_{ij}/\partial x_j = 0$ together with $\partial u/\partial x + \partial w/\partial y = 0$.

This does not make the problem steady. The Stokes equations are quasi-static in the sense that the momentum balance carries no time derivative, so the velocity field at each instant is fixed by the instantaneous geometry; the geometry itself evolves through the kinematic condition in (21), which retains $\partial h/\partial t$. The time dependence migrates entirely into the free-surface evolution, and the sheet remains history-dependent. What follows keeps that dependence throughout: the inconsistency demonstrated below arises with the domain motion fully retained, and is not a consequence of neglecting it.

Carrying the same slender expansion through, the reduced system is

$$\underbrace{\frac{\partial h}{\partial t} + \frac{\partial}{\partial x}(vh) = 0}_{\text{unchanged; time retained}}, \qquad \underbrace{0 = 4\mu\frac{\partial}{\partial x}\left(h\frac{\partial v}{\partial x}\right) + \gamma h\frac{\partial^3 h}{\partial x^3}}_{\text{quasi-static}}. \tag{28}$$

Neglecting the capillary term by (4) and integrating the momentum equation once in x, the viscous stress $4\mu h\,\partial_x v$ is independent of x. Its value is set by the edge condition (3), giving

$$4\mu\, h\frac{\partial v}{\partial x} = -\gamma \qquad \text{for all } x. \tag{29}$$

Combining (29) with the Lagrangian form of (1), $\mathrm{D}h/\mathrm{D}t = -h\,\partial_x v$, gives

$$\frac{\mathrm{D}h}{\mathrm{D}t} = \frac{\gamma}{4\mu} \qquad \Longleftrightarrow \qquad \frac{\mathrm{D}H}{\mathrm{D}T} = 1 \quad \text{everywhere}, \tag{30}$$

so *every* material element thickens at the same constant rate and $H = 1 + T$ throughout the sheet.

This is inconsistent. The sheet is initially flat and at rest, and for a long or semi-infinite sheet the far field must remain flat and quiescent, $H \to 1$ and $V \to 0$; (30) instead thickens the whole sheet uniformly and instantaneously, no matter how distant. The purely viscous solution cannot

satisfy the initial and far-field conditions, and the resolution is that inertia, however small, is what localises the disturbance near the retracting edge and keeps the problem well posed [1]. Balancing the inertial and viscous terms of (2) is then not optional, and it produces the scales (5).

Contrast (30) with the correct result of §1.3: there $H(X_c, T) = 1 + T$ holds *at the tip*, with H relaxing to unity away from it, and that spatial structure is exactly what inertia supplies.

The failure is specific to the planar geometry. As noted in [1], in the axisymmetric problem of an expanding hole the viscous operator becomes $\partial_r\big(hr^{-1}\partial_r(rv_r)\big)$, for which a purely viscous solution *is* compatible with the initial and boundary conditions, and is so regardless of the value of l_0/h_0. The necessity of retaining inertia at large Oh_p is therefore a property of this configuration, and of its reduced dimensionality, rather than a general feature of viscous sheet retraction.

B Fidelity of the Ahsan *et al.* [1] code transcription

ABDS-CN is a transcription of released MATLAB into Python. To establish that the transcription is faithful rather than merely plausible, the original code was executed under GNU Octave and compared with the transcription at identical parameters. Over six configurations the largest absolute difference in the retraction speed is 5.0×10^{-10}, and the largest normalised difference across all twenty-four recorded quantities is 3.0×10^{-8}. The former is the resolution at which the original code writes its output, so the two agree to every digit that can be compared.

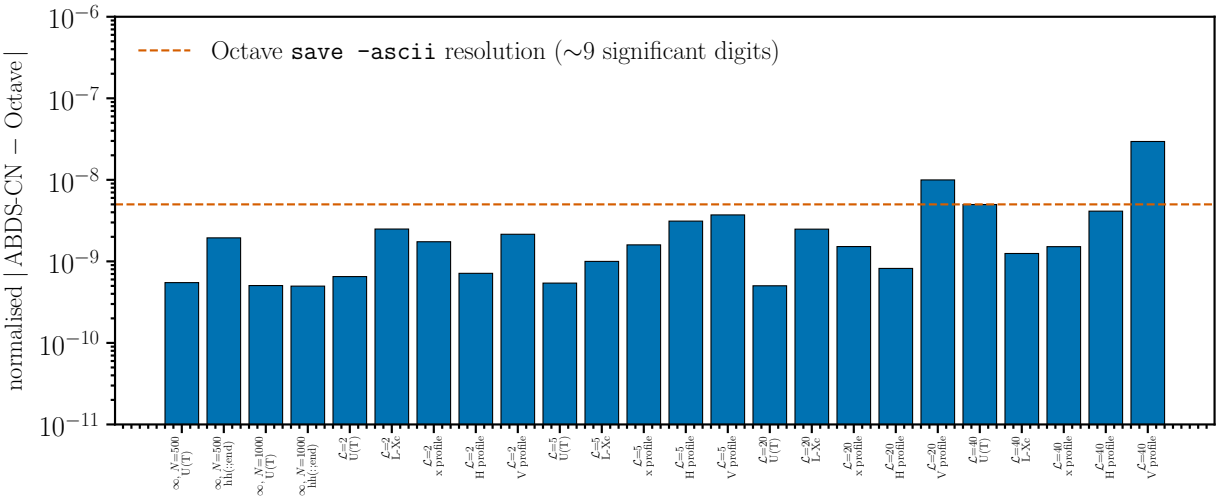


Figure 6: Original released code executed under GNU Octave, against the Python transcription ABDS-CN, at several $\mathcal{L}$.

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